Developing a Reporting Guideline for Social and Psychological Intervention Trials

Understanding randomized controlled trials of complex social and psychological interventions requires a detailed description of the interventions tested and the methods used to evaluate them. However, randomized controlled trial reports often omit, or inadequately report, this information.

Incomplete and inaccurate reporting hinders the optimal use of research, wastes resources, and fails to meet ethical obligations to research participants and consumers. We explain how reporting guidelines have improved the quality of reports in medicine, and describe the ongoing development of a new reporting guideline for randomized controlled trials: an extension of the Consolidated Standards of Reporting Trials for social and psychological interventions.

We invite readers to participate in the project by visiting our Web site, to help us reach the best-informed consensus on these guidelines (http://tinyurl.com/consort-study).
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SOCIAL AND PSYCHOLOGICAL interventions aim to improve physical health, mental health, and associated social outcomes. They are often complex and typically involve multiple, interacting intervention components (e.g., several behavior change techniques) that may act and target outcomes on several levels (e.g., individual, family, community).[1] Moreover, these interventions may be contextually dependent upon the hard-to-control environments in which they are delivered, (e.g., health care settings, correctional facilities).[2,3] The functions and processes of these interventions may be designed to accommodate particular individuals or contexts, taking on different forms while still aiming to achieve the same objective.[4,5]

Complex interventions are common in public health, psychology, education, social work, criminology, and related disciplines. For example, multisystemic therapy is an intensive intervention for juvenile offenders. Based on social-ecological and family systems theories, multisystemic therapy providers target a variety of individual, family, school, peer, neighborhood, and community influences on psychosocial and behavioral problems.[6] Treatment teams of professional therapists and caseworkers work with individuals, their families, and their peer groups to provide tailored services.[7] These services may be delivered in homes, social care, and community settings. Other examples of social and psychological interventions may be found in reviews by the Cochrane Collaboration (e.g., the Developmental, Psychosocial, and Learning Problems Group, the Cochrane Public Health Group) and the Campbell Collaboration.[8,9]

To understand their effects and to keep services up to date, academics, policymakers, journalists, clinicians, and consumers rely on research reports of intervention studies in scientific journals. Such reports should explain the methods, including the design, delivery, uptake, and context of interventions, as well as subsequent results. Accurate, complete, and transparent reporting is essential for readers to make best use of new evidence, to achieve returns on research investment, to meet ethical obligations to research participants and consumers of interventions, and to minimize waste in research. However, randomized controlled trials are often poorly reported within and across disciplines including criminology,[10] social work,[11] education,[12] psychology,[13,14] and public health.[15] Biomedical researchers have developed guidelines to improve the reporting of randomized controlled trials of health-related interventions.[16] However, many social and behavioral scientists have not fully adopted these guidelines, which may not be wholly adequate for social and psychological interventions in their current form.[10,13,17,18] Because of the unique features of these interventions, updated reporting guidance is needed.

We describe the development of a reporting guideline that aims to improve the quality of reports of randomized controlled trials of social and psychological interventions. We explain how reporting guidelines have improved the quality of reports in medicine, and why guidelines have not yet improved the quality of reports in other disciplines. We introduce a plan to develop a new reporting guideline for randomized controlled trials—an extension of Consolidated Standards of Reporting Trials (CONSORT) for social and psychological interventions—which will be written by using recommended techniques for guideline development and dissemination.[19] Wide stakeholder involvement and consensus are needed to create a useful, acceptable, and evidence-based guideline, so we hope to recruit stakeholders from multiple disciplines and professions.

Randomized trials are not the only rigorous method for evaluating interventions; many alternatives exist when randomized controlled trials are not possible or appropriate because of scientific, practical, and ethical concerns.[20] Nonetheless, randomized controlled trials are important to policymakers, practitioners, scientists, and service users, as they are generally considered the most valid and reliable research method for estimating the effectiveness of interventions.[21] Moreover, many of the issues faced in reporting randomized controlled trials also relate to other evaluation designs. As a result, this project will focus on standards for randomized controlled trials that could then also inform the development of future guidelines for other evaluation designs.

Reporting guidelines list (in the form of a checklist) the minimum information required to understand the methods and results of studies. They do not prescribe research conduct, but facilitate the writing of transparent reports by authors and appraisal of reports by research consumers. For example, the CONSORT Statement 2010 is an evidence-based guideline; to identify items, the developers reviewed evidence of trial design and conduct that could contribute to bias. With consensus methods, they developed a checklist of 25 items and a flow diagram.[16] CONSORT has improved the reporting of thousands of medical experiments.[22] It has been endorsed by more than 600 journals,[23] and it is supported by the Institute of Educational Sciences.[12] CONSORT is the only guideline for reporting randomized controlled trials that has been developed with such rigor, and it has remained more prominent than any other guideline for more than 15 years; for greatest impact, then, any further reporting guidelines related to randomized controlled trials should be developed in collaboration with the CONSORT Group.

LIMITATIONS OF PREVIOUS REPORTING GUIDELINES

Researchers and journal editors in the social and behavioral sciences are generally aware of CONSORT but often object that it is not fully appropriate for social and psychological interventions.[10,13,17,18] As a result, uptake of CONSORT guidelines in these disciplines is low. Although some criticisms are attributable to inaccurate perceptions about common features of randomized controlled trials across disciplines, many relate to real limitations for social and psychological interventions.[24] For example, CONSORT is most relevant to randomized controlled trials in medical disciplines; it was developed by biostatisticians and medical researchers with minimal input from experts in other disciplines. Journal editors, as well as social and behavioral science researchers, believe there is a need to include appropriate stakeholders in developing a new, targeted guideline to improve uptake in their disciplines.[12,25]

The CONSORT Group has produced extensions of the original CONSORT Statement relevant to social and psychological interventions, such as additional checklists for cluster,[26] nonpharmacological,[27] pragmatic,[28] and quality-of-life randomized controlled trials.[29] These extensions provide important insights, but complex social and psychological interventions, for example, include multiple, interacting components at several levels, with various outcomes. These randomized controlled trials require use of several extensions at once, creating a barrier to guideline uptake; increasing intervention complexity also gives rise to new issues that are not included in existing guidelines. Therefore, simply disseminating CONSORT guidelines as they stand is insufficient because this would not address the need for editors and authors to "buy-in" to this process. To improve uptake in these disciplines, CONSORT guidelines need to be extended to specifically address the important features of social and psychological interventions.

Social and behavioral scientists have developed other reporting guidelines, including the Workgroup for Intervention Development and Evaluation Research Recommendations for behavioral change interventions,[14,30] the American Educational Research Association's Standards for Reporting Research,[31] the Reporting of Primary Empirical Research Studies in Education Guidelines,[32] and the Journal Article Reporting Standards (JARS) of the American Psychological Association (APA).[33] Although they address issues not covered by the CONSORT Statement and its extensions, these guidelines (except for JARS[33]) do not provide specific guidance for randomized controlled trials. Moreover, compared with the CONSORT Statement and its official extensions, guidelines in the social and behavioral sciences have not consistently followed optimal techniques for guideline development and dissemination that are recommended by international leaders in the advancement of reporting guidelines,[19] such as the use of systematic literature reviews and formal consensus methods to select reporting standards.[34] Researchers in public health, psychology, education, social work, and criminology have noted that these guidelines could be more user-friendly and dissemination could benefit from up-to-date knowledge-transfer techniques.[11–13,17,30,35,36]

For example, JARS—a notable and valuable guideline for empirical psychological research—is endorsed by few journals outside the APA, whereas CONSORT is endorsed by hundreds of journals internationally. According to the Institute for Scientific Information Web of Knowledge and Google Scholar citations, JARS is cited approximately a dozen times annually, whereas CONSORT guidelines are cited hundreds of times per year. Moreover, the APA commissioned a select group of APA journal editors and reviewers to develop JARS, and the group based most of their work on existent CONSORT guidelines; by comparison, official CONSORT extensions have been developed by using rigorous consensus methods, have involved various international stakeholders in guideline development and dissemination, and update content on the most recent scientific literature. Nonetheless, no current CONSORT guideline adequately addresses the unique features of social and psychological interventions. This new CONSORT extension will incorporate lessons from previous extensions, reporting guidelines, and the research literature to aid the critical appraisal, replication, and uptake of this research.

ASPECTS OF INTERNAL VALIDITY

Internal validity is the extent to which the results of a study may be influenced by bias. Like other study designs, the validity of randomized controlled trials depends on high-quality execution. Poorly conducted randomized controlled trials can produce more biased results than well-conducted randomized controlled trials and well-conducted nonrandomized studies.[37,38] For example, evidence indicates that randomized controlled trials that do not adequately conceal the randomization sequence can exaggerate effect estimates by up to 30%,[39] and low-quality reports of these randomized controlled trials are associated with effect estimates exaggerated by up to 35%.[40] Social and psychological intervention randomized controlled trials are susceptible to these risks of bias as well.

Some aspects of internal validity, although included in CONSORT, remain poorly reported—even in the least complex social and psychological intervention studies. Reports of randomized controlled trials should describe procedures for minimizing selection bias, but reports often omit information about random sequence generation and allocation concealment,[35,41] and psychological journals report methods of sequence generation less frequently than medical journals.[13] A review of educational reports found no studies that adequately reported allocation concealment,[12] and reports in criminology often lack information about randomization procedures.[10,25] Randomized controlled trials of social and psychological interventions may also use nontraditional randomization techniques, such as stepped-wedge or natural allocation,[42] which need to be thoroughly described. In addition, reports of social and psychological intervention trials often fail to include details about trial registration, protocols, and adverse events,[35,41] which may include important negative consequences at individual, familial, and community levels.

Other aspects of CONSORT may require greater emphasis or modification for randomized controlled trials of social and psychological interventions. In developing this CONSORT extension, we expect to identify new items and to adapt existing items that relate to the internal validity. These may include items discussed during the development of previous CONSORT extensions or other guidelines, as well as items suggested by participants in this project. For example, it may not be possible to blind participants and providers of interventions, but blinding of outcome assessors is often possible but rarely reported, and few studies explain if blinding was maintained or how lack of blinding was handled.[17,35,41] In social and psychological intervention studies, outcome measures are often subjective, variables may relate to latent constructs, and information may come from multiple sources (e.g., participants, providers). Though an issue in other areas of research, the influence on randomized controlled trial results of the quality of subjective outcome measures in social and psychological intervention research has long been highlighted because of their prevalence in this type of research.[43] Descriptions of the validity, reliability, and psychometric properties of such measures are therefore particularly useful for social and psychological intervention trials, especially when they are not widely available or discussed in the research literature.[26,44] Moreover, multiple measures may be analyzed in several ways, so authors need to transparently report which procedures were performed and to explain their rationale.

ASPECTS OF EXTERNAL VALIDITY

External validity is the extent to which a study's results are applicable in other settings or populations. Currently, as randomized controlled trials are primarily designed to increase the internal validity of study findings, the CONSORT Statement gives relatively little attention to external validity. Although high internal validity is an important precondition for any discussion of a randomized controlled trial's external validity, updating the CONSORT Statement to include more information about external validity is critical for the relevance and uptake of a CONSORT extension for social and psychological interventions. These interventions may be influenced by context, as different underlying social, institutional, psychological, and physical structures may yield different causal and probabilistic relations between interventions and observed outcomes. Contextual information is necessary to compare the effectiveness of an intervention across time and place.[45] Lack of information relevant to external validity may prevent practitioners or policymakers from using evidence appropriately to inform decision-making, yet existing guidelines do not adequately explain how authors should describe (1) how interventions work, (2) for whom, and (3) under what conditions.[46]

First, it is useful for authors to explain the key components of interventions, how those components could be delivered, and how they relate to the outcomes selected. At present, authors can follow current standards for reporting interventions without providing adequate details about complex interventions.[47] Many reports do not contain sufficient information about the interventions tested, and many reports also do not reference treatment manuals.[48] Providing logic models—as described in the Medical Research Council Framework for Complex Interventions[49]—or presenting theories of change can help elucidate links in causal chains that can be tested, identify important mediators and moderators, and facilitate syntheses in reviews.[50] Moreover, interventions are rarely implemented exactly as designed, and complex interventions may be designed to be implemented with some flexibility to accommodate differences across participants,[4] so it is important to report how interventions were actually delivered by providers and actually received by participants.[51] Particularly for social and psychological interventions, the integrity of implementing the intended functions and processes of the intervention are essential to understand.[4] As randomized controlled trials of a particular intervention can yield different relative effects depending on the nature of the control groups, information about delivery and uptake should be provided for all trial arms.[52]

Second, reports should describe recruitment processes and representativeness of samples. Participants in randomized controlled trials of social and psychological interventions are often recruited outside routine practice settings via processes that differ from routine services.[31] An intervention that works for one group of people may not work for people living in different cultures or physical spaces, or it may not work for people with slightly different problems and comorbidities. Enrolling in a randomized controlled trial can be a complex process that affects the measured and unmeasured characteristics of participants, and recruitment may differ from how users normally access interventions. Well-described randomized controlled trial reports will include the characteristics of all participants (volunteers, those who enrolled, and those who completed) in sufficient detail for readers to assess the comparability of the study sample to populations and in everyday services.[31,33,53]

Finally, as these interventions often occur in social environments, reports should describe factors of the randomized controlled trial context that are believed to support, attenuate, or frustrate observed effects.[54] Interventions may differ across groups of different social or socioeconomic positions, and equity considerations should be addressed explicitly.[55,56] Several aspects of setting and implementation may be important to consider, such as administrative support, staff training and supervision, organizational resources, the wider service system, and concurrent political or social events.[5,47,57,58] Reporting process evaluations may help readers understand mechanisms and outcomes.

DEVELOPING A NEW CONSORT EXTENSION

This new reporting guideline for randomized controlled trials of social and psychological interventions will be an official extension of the CONSORT Statement. Optimally, it will help improve the reporting of these studies. Like other official CONSORT extensions,[26,28,59,60] this guideline will be integrated with the CONSORT Statement and previous extensions, and updates of the CONSORT Statement may incorporate references to this extension.

The project is being led by an international collaboration of researchers, methodologists, guideline developers, funders, service providers, journal editors, and consumer advocacy groups. We will be recruiting participants in a manner similar to that of other reporting guideline initiatives—identifying stakeholders through literature reviews, the project's International Advisory Group, and stakeholder-initiated interest in the project.[14,16] We hope to recruit stakeholders with expertise from all related disciplines and regions of the world, including low- and middle-income countries. Methodologists will identify items that relate to known sources of bias, and they will identify items that facilitate systematic reviews and research synthesis. Funders will consider how the guideline can aid the assessment of grant applications for randomized controlled trials and methodological innovations in intervention evaluation. Practitioners will identify information that can aid decision-making. Journal editors will identify practical steps to implement the guideline and to ensure uptake.

We will use consensus techniques to reduce bias in group decision-making and to promote widespread guideline uptake and knowledge translation activities upon project completion.[61] Following rigorous reviews of existing guidelines and current reporting quality, we will conduct a Delphi process to identify a prioritized list of reporting items to consider for the extension. That is, we will invite a group of experts to answer questions about reporting items and to suggest further questions. We will circulate their feedback to the group and ask a second round of questions. The Delphi process will capture a variety of international perspectives and allow participants to share their views anonymously. Following the Delphi process, we will host a consensus meeting to review the findings and to generate a list of minimal reporting standards, mirroring the development of previous CONSORT guidelines.[16,27,28]

Together, participants in this process will create a checklist of reporting items and a flowchart for reporting social and psychological intervention randomized controlled trials. In addition, we will develop an explanation and elaboration document to explain the scientific rationale for each recommendation and to provide examples of clear reporting; a similar document was developed by the CONSORT Group to help disseminate a better understanding for each included checklist item.[62] This document will help persuade editors, authors, and funders of the importance of the guideline. It will be a useful pedagogical tool, helping students and researchers understand the methods for conducting randomized controlled trials of social and psychological interventions, and it will help authors meet the guideline requirements.[19]

The success of this project depends on widespread involvement and agreement among key international stakeholders in research, policy, and practice. For example, previous developers have obtained guideline endorsement by journal editors who require authors and peer reviewers to use the guideline during manuscript submission and who must enforce journal article word limits.[19,63] Many journal editors have already agreed to participate, and we hope other researchers and stakeholders will volunteer their time and expertise.

CONCLUSIONS

Reporting guidelines help us use scarce resources efficiently and ethically. Randomized controlled trials are expensive, and the public has a right to expect returns on their investments through transparent, usable reports. When randomized controlled trial reports cannot be used (for whatever reason), resources are wasted. Participants contribute their time and put themselves at risk for harm to generate evidence that will help others, and researchers should disseminate that information effectively.[17] Policymakers benefit from research when developing effective, affordable standards of practice and choosing which programs and services to fund. Administrators and managers are required to make contextually appropriate decisions. Transparent reporting of primary studies is essential for their inclusion in systematic reviews that inform these activities. For example, there is the need to determine if primary studies are comparable, examine biases within included studies, assess the generalizability of results, and implement effective interventions. Finally, we hope this guideline will reduce the effort and time required for authors to write reports of randomized controlled trials.

Randomized controlled trials are not the only valid method for evaluating interventions,[20] nor are they the only type of research that would benefit from better reporting.[64] Colleagues have identified the importance of reporting standards for other types of research, including observational,[65] quasi-experimental,[66] and qualitative studies.[67] This guideline is the first step toward improving reports of many designs for evaluating social and psychological interventions that we hope will be addressed by this and future projects. We invite stakeholders from disciplines that frequently research these interventions to join this important effort and participate in guideline development by visiting our Web site where they can find more information about the project, see updates on its progress, and sign up to be involved (http://tinyurl.com/consort-study). ■

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P. Montgomery, E. Mayo-Wilson, and S. Grant conceptualized the idea for the project. All authors helped draft the article, and all have read and approved the final article.

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